

Weber Electrodynamics

Constants and units

We use absolute system of units throughout this document, unless otherwise specified.

	Description	SI Unit	Gauss-Weber absolute units
q^2	electric charge squared	C^2	$\frac{mg \cdot mm^3}{s^2}$
ϵ_0	permittivity of free space	$\frac{C^2 \cdot s^2}{kg \cdot m^3}$	dimensionless
c	Meters per second (speed of light)	$\frac{m}{s}$	$\frac{mm}{s}$
q_i	electric charge of particle i	C	$\sqrt{\frac{mg \cdot mm^3}{s^2}}$
m_i	mass of particle i	kg	mg

2-Particle system notation

Mass and Charge of Particle 1 and Particle 2

$$m_1, q_1$$

$$m_2, q_2$$

Positions for Particle 1 and Particle 2

Position

$$x_1(t), y_1(t), z_1(t)$$

$$x_2(t), y_2(t), z_2(t)$$

Velocity

$$\dot{x}_1(t), \dot{y}_1(t), \dot{z}_1(t)$$

$$\dot{x}_2(t), \dot{y}_2(t), \dot{z}_2(t)$$

Acceleration

$$\ddot{x}_1(t), \ddot{y}_1(t), \ddot{z}_1(t)$$

$$\ddot{x}_2(t), \ddot{y}_2(t), \ddot{z}_2(t)$$

Vector notation for Particle 1 and Particle 2

Position

$$\vec{r}_1 = (x_1(t), y_1(t), z_1(t)), \quad \vec{r}_2 = (x_2(t), y_2(t), z_2(t))$$

Velocity

$$\vec{v}_1 = (\dot{x}_1(t), \dot{y}_1(t), \dot{z}_1(t)), \quad \vec{v}_2 = (\dot{x}_2(t), \dot{y}_2(t), \dot{z}_2(t))$$

Acceleration

$$\vec{a}_1 = (\ddot{x}_1(t), \ddot{y}_1(t), \ddot{z}_1(t)), \quad \vec{a}_2 = (\ddot{x}_2(t), \ddot{y}_2(t), \ddot{z}_2(t))$$

Momenta for Particle 1 and Particle 2

$$p_{x_1} = m_1 \dot{x}_1$$

$$p_{y_1} = m_1 \dot{y}_1$$

$$p_{z_1} = m_1 \dot{z}_1$$

$$p_{x_2} = m_2 \dot{x}_2$$

$$p_{y_2} = m_2 \dot{y}_2$$

$$p_{z_2} = m_2 \dot{z}_2$$

Relative Coordinates and Vectors for Particle 1 and Particle 2

Relative position

$$x = x_1 - x_2$$

$$y = y_1 - y_2$$

$$z = z_1 - z_2$$

Relative velocity

$$\dot{x} = \dot{x}_1 - \dot{x}_2$$

$$\dot{y} = \dot{y}_1 - \dot{y}_2$$

$$\dot{z} = \dot{z}_1 - \dot{z}_2$$

Relative acceleration

$$\ddot{x} = \ddot{x}_1 - \ddot{x}_2$$

$$\ddot{y} = \ddot{y}_1 - \ddot{y}_2$$

$$\ddot{z} = \ddot{z}_1 - \ddot{z}_2$$

Relative vectors

$$\vec{r} = (x, y, z)$$

$$\vec{v} = (\dot{x}, \dot{y}, \dot{z})$$

$$\vec{a} = (\ddot{x}, \ddot{y}, \ddot{z})$$

Relative distance

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

Relative radial velocity

$$\dot{r} = \frac{x\dot{x} + y\dot{y} + z\dot{z}}{r} = \frac{\vec{r} \cdot \vec{v}}{r}$$

Relative radial acceleration

$$\begin{aligned}\ddot{r} &= \frac{x\ddot{x} + y\ddot{y} + z\ddot{z}}{r} + \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{r} - \frac{(x\dot{x} + y\dot{y} + z\dot{z})^2}{r^3} \\ &= \frac{\vec{r} \cdot \vec{a}}{r} + \frac{\vec{v} \cdot \vec{v}}{r} - \frac{(\vec{r} \cdot \vec{v})^2}{r^3} \\ &= \frac{1}{r} (\vec{r} \cdot \vec{a} + \vec{v} \cdot \vec{v} - (\hat{r} \cdot \vec{v})^2)\end{aligned}$$

Unit relative position vector

$$\hat{r} = \frac{\vec{r}}{r}$$

Weber Potential Energy

For a two-particle system, we denote the distance between particles 1 and 2 as r_{12} . To keep expressions clean, we write $r = r_{12}$, $\dot{r} = \dot{r}_{12}$, $\ddot{r} = \ddot{r}_{12}$ when the context is clear.

In SI units:

$$U_{SI} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \left(1 - \frac{\dot{r}^2}{2c^2} \right)$$

In absolute units:

$$U = \frac{q_1 q_2}{r} \left(1 - \frac{\dot{r}^2}{2c^2} \right)$$

Weber Force

Explicit Time Parameterization of r and $\dot{r}$

If r and $\dot{r}$ are both functions of time, that is, $r = r(t)$ and $\dot{r} = \dot{r}(t)$, then the force on particle 1 due to particle 2 can be derived from the potential energy using $\vec{F}_{12} = -\nabla_1 U$. Since $\nabla_1 r = \hat{r}$ (where $\hat{r}$ points from particle 2 to particle 1), we have:

$$\vec{F}_{12} = -\frac{dU}{dr} \hat{r}$$

Using the chain rule:

$$\frac{d\dot{r}^2}{dr} = 2\dot{r} \frac{d\dot{r}}{dr} = 2\dot{r} \frac{d\dot{r}/dt}{dr/dt} = 2\dot{r} \frac{\ddot{r}}{\dot{r}} = 2\ddot{r}$$

Computing the derivative:

$$\frac{dU}{dr} = -\frac{q_1 q_2}{r^2} \left(1 - \frac{\dot{r}^2}{2c^2} + \frac{r\ddot{r}}{c^2} \right)$$

This yields:

$$\vec{F}_{12} = \frac{q_1 q_2}{r^2} \hat{r} \left(1 - \frac{\dot{r}^2}{2c^2} + \frac{r\ddot{r}}{c^2} \right)$$

Implicit coordinate-based formulation of r and $\dot{r}$

If we take ∇_1 as the gradient operator with respect to the coordinates of particle 1, that is $\nabla_1 = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}, \frac{\partial}{\partial z_1} \right)$, then we can derive Weber's force as follows:

$$\vec{F}_{12} = -\nabla_1 U$$

where the first component of the force is given by:

$$-\frac{\partial U}{\partial x_1} = q_1 q_2 \frac{(x_1 - x_2)}{r^3} \left(1 - \frac{\dot{r}^2}{2c^2} + \frac{r\ddot{r}}{c^2} \right)$$

Weber's force can be rewritten in vector form as:

$$\vec{F}_{12} = \frac{q_1 q_2}{r^2} \hat{r} \left(1 + \frac{1}{c^2} \left(\vec{v} \cdot \vec{v} + \vec{r} \cdot \vec{a} - \frac{3}{2} (\hat{r} \cdot \vec{v})^2 \right) \right)$$

Lagrangian formulation

$$T = \frac{1}{2} m_1 (\vec{v}_1 \cdot \vec{v}_1) + \frac{1}{2} m_2 (\vec{v}_2 \cdot \vec{v}_2)$$

We define S as the velocity-dependent interaction term with a positive sign in the velocity-dependent term (note this differs from the potential energy U which has a negative sign). Again, we write $r = r_{12}$ and $\dot{r} = \dot{r}_{12}$ for cleaner notation:

$$S = \frac{q_1 q_2}{r} \left(1 + \frac{\dot{r}^2}{2c^2} \right)$$

$$L = T - S$$

In the following, we use q_i to denote generalized coordinates. For a 2-particle system in 3D:

$$\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \\ \dot{q}_5 \\ \dot{q}_6 \end{pmatrix} = \begin{pmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{z}_1 \\ \dot{x}_2 \\ \dot{y}_2 \\ \dot{z}_2 \end{pmatrix}$$

The generalized coordinates should not be confused with the electric charges. The Hamiltonian is defined as:

$$H = \left(\sum_i^6 \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) - L$$

Performing the algebraic expansion yields:

$$H = T + U$$

Hamiltonian formulation

For an n-body system of particles interacting via Weber electrodynamics, the kinetic energy for each particle is

$$T_i = \frac{1}{2} m_i (\vec{v}_i \cdot \vec{v}_i)$$

The potential energy for a pair of particles is given by (in SI units):

$$U_{ij} = \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}} \left(1 - \frac{\dot{r}_{ij}^2}{2c^2} \right)$$

where r_{ij} is the distance between particles i and j , and $\dot{r}_{ij}$ is its time derivative. Here q_i and q_j denote the electric charges of particles i and j .

The Hamiltonian for n charges is then given by

$$H = \sum_i^n T_i + \sum_{i<j}^n U_{ij}$$

Euler-Lagrange equations

The Euler-Lagrange equation for Weber electrodynamics yields the force on particle 1 in the x -direction:

$$F_{12}^x = \frac{d}{dt} \frac{\partial S}{\partial \dot{x}_1} - \frac{\partial S}{\partial x_1} = q_1 q_2 \frac{(x_1 - x_2)}{r^3} \left(1 - \frac{\dot{r}^2}{2c^2} + \frac{r\ddot{r}}{c^2} \right)$$

with analogous equations for the y and z directions and for particle 2.

Equations of motion

Hamilton's equations give the time evolution of positions and momenta:

$$\dot{x}_i = \frac{\partial H}{\partial p_{x_i}}, \quad \dot{p}_{x_i} = -\frac{\partial H}{\partial x_i}$$

For the two-particle Weber system, these expand to:

$$\dot{x}_1 = \frac{1}{m_1} \left(p_{x_1} + \frac{q_1 q_2}{c^2} \frac{\dot{r}(x_1 - x_2)}{r^2} \right)$$

$$\dot{x}_2 = \frac{1}{m_2} \left(p_{x_2} - \frac{q_1 q_2}{c^2} \frac{\dot{r}(x_1 - x_2)}{r^2} \right)$$

$$\dot{p}_{x_1} = \frac{q_1 q_2}{r^2} \left[\frac{(x_1 - x_2)}{r} \left(1 - \frac{3\dot{r}^2}{2c^2} \right) + \frac{\dot{r}(\dot{x}_1 - \dot{x}_2)}{c^2} \right]$$

$$\dot{p}_{x_2} = -\dot{p}_{x_1}$$

with analogous equations for the y and z directions.